An elementary obstruction for $y^3 + y^2 + xy = x^4 + 1$

A self-contained non-existence proof

Theorem. *There is no pair $(x, y) \in \mathbb{Z}^2$ satisfying*

$$y^3 + y^2 + xy = x^4 + 1. \tag{1}$$

An elementary divisor observation.

Lemma. *Let $u \in \mathbb{Z}$. Every positive divisor d of $u^2 + u + 1$ with $3 \nmid d$ satisfies $d \equiv 1 \pmod 3$.*

Proof. Let $p \neq 3$ be a prime divisor of $u^2 + u + 1$. Modulo p, we have $u^3 = 1$ and $u \neq 1$: the latter follows because substitution of $u = 1$ would give $p \mid 3$. Multiplication by u therefore partitions the $p - 1$ nonzero residue classes into disjoint triples $\{r, ur, u^2 r\}$. These three classes are distinct, since $u^3 = 1$ and $u \neq 1$. Hence $3 \mid p - 1$. Every prime factor of d is consequently $1 \pmod 3$, and so is d; this includes the empty product $d = 1$. □

Proof of the theorem. Suppose (1) holds. Reduction modulo 3 gives

$$0 \equiv y^2 + (x+1)y - x^2 - 1 \equiv (x-1)^2 + (x - y + 1)^2 \pmod 3.$$

The only squares modulo 3 are 0 and 1, so both squares vanish. Thus

$$x \equiv 1, \qquad y \equiv 2 \pmod 3. \tag{2}$$

Define

$$A = x^2 + y, \qquad B = x^4 - x^2 y + y^2 - x^2 + y + x.$$

A direct expansion, followed by (1), yields

$$AB = x^6 + x^3 + 1 = \left(x^3 + \frac{1}{2}\right)^2 + \frac{3}{4} > 0. \tag{3}$$

In particular $A \neq 0$. If $A < 0$, then $t = -y \geq x^2 + 1$ and

$$y^2 + y + x = t(t-1) + x \geq x^2(x^2+1) + x = x^4 + x(x+1) \geq 0,$$

where $x(x+1) \geq 0$ for every integer x. Since $y < 0$, this contradicts $y(y^2 + y + x) = x^4 + 1 > 0$. Therefore $A > 0$, and (3) also gives $B > 0$.

On the other hand, (2) implies

$$B \equiv 1 - 2 + 1 - 1 + 2 + 1 \equiv 2 \pmod 3.$$

But B is a positive divisor of $(x^3)^2 + x^3 + 1$ not divisible by 3. The lemma, with $u = x^3$, forces $B \equiv 1 \pmod 3$, a contradiction. □

How the obstruction is found

Make two terms cancel before choosing a modulus. Write

$$F(x,y) = y^3 + y^2 + xy - x^4 - 1.$$

The terms y^2 and x^4 suggest substituting $y = -x^2$. This cancels those two terms exactly and leaves

$$F(x,-x^2) = -(x^6 + x^3 + 1).$$

Consequently, an integer zero of F must make $x^2 + y$ a divisor of $x^6 + x^3 + 1$. The useful modulus is therefore not a fixed small integer: it is the expression $x^2 + y$ suggested by the equation itself.

Recognize the restriction on prime divisors. The remaining expression is

$$x^6 + x^3 + 1 = u^2 + u + 1, \qquad u = x^3.$$

A prime divisor other than 3 must admit a nontrivial cube root of unity, which is exactly what the triples in the lemma express. It must therefore be $1 \pmod 3$. This turns the substitution into a restriction on *every positive divisor*, rather than merely a congruence on x or y.

Use the complementary factor. The original equation modulo 3 forces $(x,y) \equiv (1,-1)$, so $x^2 + y \equiv 0 \pmod 3$. The exceptional prime 3 thus obscures the obstruction in this first factor. Polynomial division isolates the complementary factor B through the exact identity

$$F(x,y) = (x^2+y)(x^4 - x^2y + y^2 - x^2 + y + x) - (x^6 + x^3 + 1).$$

Unlike $x^2 + y$, this complementary factor is $-1 \pmod 3$, so it is coprime to 3 and directly contradicts the lemma once it is positive.

Do not omit the sign check. A negative divisor can be $2 \pmod 3$ without having any prime factor $2 \pmod 3$; for example, -1 does this. Thus the positivity argument is essential. The equation itself rules out $x^2 + y < 0$, and the positive product in (3) then forces $B > 0$. No sign assumption on the unknown integers is being made.

The proof combines the rigid residue pair modulo 3 with a prime-divisor restriction exposed by cancellation. It excludes every integer pair, without a search bound or an unproved auxiliary claim.